

Significance

Craniopharyngiomas (CPs) are rare (incidence of 0.5-2 cases per million people annually), benign yet locally aggressive tumors of the sellar and parasellar regions, with a low histological grade, located near the optic chiasm, hypothalamus, and third ventricle. They primarily affect children aged 5-14 years, with 30%–50% of all cases occurring during childhood and adolescence and are also found in adults aged 50-74 years ([1]).

There are two histologic subtypes: adamantinomatous (ACP) and papillary (PCP). Distinguishing them is clinically important: ACP typically has higher recurrence and worse prognosis, reflecting more infiltrative growth that can compromise vision and hypothalamic–pituitary function, leading to endocrine deficits, visual disturbances, decreased height, obesity (hypothalamic obesity), strokes and cognitive complications ([3]).

The diagnosis of CP is often delayed for years, as initial symptoms include intracranial pressure. Additionally, visual impairment (62%–84%) and endocrine deficits (52%–87%) are common. Endocrine dysfunction results from disruption of the hypothalamic–pituitary axis, affecting growth hormone (75%), gonadotropins (40%), adrenocorticotrophic hormone (25%), and thyroid-stimulating hormone (25%); overall, 40%–87% of patients have at least one hormonal deficiency at diagnosis ([4], [5]). CTs and MRIs are also frequently diagnostic for pediatric CP, with most tumors showing intratumoral calcifications and cystic components.

Even though overall survival rates are high (92%), recurrences after complete resection and progressions after incomplete resection can be expected in the range of 71-90%, with radiotherapy reducing it to 21% ([6]).

For surgical planning, various grading systems have been developed based on MRI images and the location of anatomical structures involved with or surrounding the tumors. Due to the anatomical differences among tumors, there is no single preferred approach ([7]). As an example, one recent classification system is the quantitative sensory testing (QST)([8]), which classifies CPs into:

- **Type Q:** CPs that originate from the space below the diaphragm, with an enlarged pituitary fossa
- **Type S:** CPs that originate from the pituitary stalk and tend to spread through cisterns.
- **Type T:** CPs that originate in the top of the pars tuberalis and extend upward to occupy the third ventricle.

Because the CP tends to stick to the nearby optic pathway and surrounding tissues, removal becomes difficult, increasing the risk of complications and death. Some surgeons then prefer partial removal followed by additional post-operative radiation. Over the past decade, there has been a shift in CP treatment from aggressive tumor removal to methods that focus on preserving function. Since most pediatric CPs are ACPs with frequent cyst formation, the treatment often involves cystic decompression through aspiration and/or intracystic chemotherapy. In this context, BRAF/MEK inhibitors have demonstrated promising results for some CP patients, including targeted therapies for ACP with CTNNB1 mutations. The

Endonasal endoscopically assisted approach (EEA) has become the standard method for treating most sellar and suprasellar CPs. However, many CPs mainly develop in the infundibulo-tuberal region of the third ventricle (3V). A significant challenge is finding a safe cleavage plane due to the strong attachment between the CP and the hypothalamus, which is especially true for PCPs, as it involves cutting through the functional 3V and poses a risk of damaging the hypothalamus. Other methods, such as through the corpus callosum or opening the lamina terminalis, are used based on preoperative MRI diagnosis of the 3V topography.

AI techniques, such as deep learning models, can provide precise segmentation of CPs and adjacent structures, including the hypothalamus and third ventricle, thereby improving preoperative planning and potentially long-term outcomes ([9]).

References

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[11] Childhood Craniopharyngioma Treatment (PDQ)—Health Professional Version